Gregory Levy

gregory.levy@economics.ox.ac.uk

Selected Research Projects

Machine Learning Theory

Neural Networks as Optimal Continuous Piecewise-Linear Approximators:

Developed fundamental theoretical framework showing that ReLU neural networks behave as asymptotically optimal continuous piecewise-linear approximators. Derived and empirically verified scaling law MSE $\propto \Gamma^{2d/(d+4)}/K^{4/d}$ where Γ is an integrated Hessian Frobenius norm drawn from free-knot linear spline theory and K is network capacity. Achieved extremely strong empirical validation: $R^2 = 0.93$ (1D), $R^2 = 0.97$ (2D), $R^2 = 0.98$ (3D) across 100+ synthetic functions. Established finite-sample convergence results showing $\alpha = 0.8$ decay rate consistent with approximation theory predictions. Provides principled approach to predicting network capacity requirements and identifying failure modes. Developing meta-learning framework for curvature estimation on real-world datasets. [In preparation for ICML 2026.]

Adaptive Linear Interpolation Deviation (aLID) for Uncertainty Quantification:

Developed novel uncertainty estimation method that adaptively combines geometric and sparsity-based uncertainty measures to detect when neural networks are likely to produce poor estimates. Applies penultimate-layer feature representations to compute deviations from linear interpolation between training points in dense regions and nearest-neighbour distances in sparse regions, with density-dependent weighting. Outperforms ensemble methods and conformal prediction at significantly lower computational cost. Applications to safe AI deployment: detecting distribution shift, predicting when model outputs are unreliable, and quantifying epistemic uncertainty. Derived theoretical bounds connecting geometric uncertainty to approximation error. [In preparation for ICML 2026.]

Empirical Political Economy

Measuring Democratic Misalignment via Preference Learning:

Constructed novel measures of citizen and politician preferences capturing policy divergence across all 435 US congressional districts and 6 policy areas. ML-augmented Multilevel Regression with Poststratification (MRP) achieved RMSE of 2.5%—7.5% for citizen preference estimation (well below typical 10% threshold). Multidimensional ideal points model achieved 95% accuracy predicting politician votes on 2.99M voting records (vs. 85%—90% in existing literature). Extending to predict political behaviour from campaign websites via NLP, comparing predicted alignment to voter choices across US, UK, and Australian elections. [MPhil thesis, distinction.]

Political Uncertainty Classification and Forecasting:

Developed LLM-based classification pipeline processing 13M historical newspaper articles (late 1700s–present) for quantifying political risk and instability. Initial prototype with optimised few-shot GPT-4o prompting; currently fine-tuning BERT models on large corpus of hand-coded articles for cost-effective scaling to additional publications and languages. Generated novel historical time series of political instability for geopolitical risk forecasting. [Forthcoming in *Handbook of Political Economy* (Acemoglu & Robinson, eds.).]

Skills

Programming Languages

Python (Advanced), R, MATLAB, Stata, GDScript

Machine Learning

Large-scale LLM deployment with multi-model architectures, optimised prompt engineering

ML libraries: scikit-learn, PyTorch, TensorFlow, Hugging Face, etc.

Large, complex dataset processing (13M+ records), causal inference, experimental design

Infrastructure

Cluster computing, Docker, API integration, parallel processing, VM deployment

Education

University of Oxford (UK)

DPhil in Economics — 2028 (expected)

Econometrics, Political Economy, Foerster Lab for AI Research (FLAIR)

MPhil in Economics (with Distinction) — 2025

University of Queensland (Australia)

Honours Degree in Economics (comparable to 1-year masters) — 2022

Synergies Consulting Prize for Best Thesis

Bachelor of Mathematics — 2021

Bachelor of Economics — 2021

Research Leadership

Co-coordinator, Machine Learning and Economics Reading Group (Oxford)

Organiser (with Max Kasy) of interdisciplinary discussion group since September 2024. Led explainability focus in Michaelmas Term 2024 and presented on mechanistic interpretability research (e.g., superposition, sparse auto-encoders, etc.).

Conferences and Workshops

Selected for Plurality Institute 'LLMs and Public Discourse' workshop (Berkeley, 2025)

Selected for Machine Learning in Economics Summer Institute (University of Chicago, 2024)

Presented at Foundations of Utility and Risk (FUR) Conference (Brisbane, 2024)

Presented at Learning, Evolution, and Games (LEG) Conference (Brisbane, 2024)

Honours, Scholarships, and Awards

Magdalen College Graduate Scholarship (Oxford) — 2023

Prize for Best Honours Thesis (UQ) — 2022

Dean's Honour Roll (UQ) — 2021

UQ Global Experiences Scholarship (UQ, for exchange at McGill University) — 2019